Diagnostic features:

1. Genophore:

- Single, Circular bacterial chromosome.
- Primitive nucleus (Nucleoid) without nuclear envelop and nucleolus.

2. Plasmid:

- Double stranded, Circular, extra genomic DNA found in cytoplasm of bacteria.
- It contains genes for fertility and antibiotic resistance.

3. Polysome:

- Ribosomes are held together by mRNA and form Polyribosome or Polysome.
- 70 S Ribosomes are attached mRNA in bacteria.

4. Mesosome:

- Localized infoldings of bacterial plasma membrane.
- It is found in the form of Vesicles, tubules and lamellae.
- It helps in respiration and binary fission.

5. Fimbriae or Pili:

- Fimbriae or Pili are hair like appendages found on the surface of the bacterial cell wall.
- It helps in conjugation hence it is called as Sex pili.

Exercise : 17

Types of stele.

Aim : To study and identify the types of stele.

Principle: Stele is a central cylinder of vascular tissues consisting of xylem, phloem, pericycle and sometimes medullary rays with pith. Stelar theory demonstrates the essential differences between the principal divisions of higher plants according to the structure of their vascular systems. It has proved of great significance in the study of plant anatomy and phytogeny.

Requirements : Model/ Photograph / picture of types of stele.

Diagnostic features:

1. Actinostele:

- Star shaped xylem core surrounded by phloem
- It is found in *Lycopodium serratum*.

2. Plectostele:

- Xylem plates alternates with Phloem plates.
- It is found in *Lycopodium clavatum*.

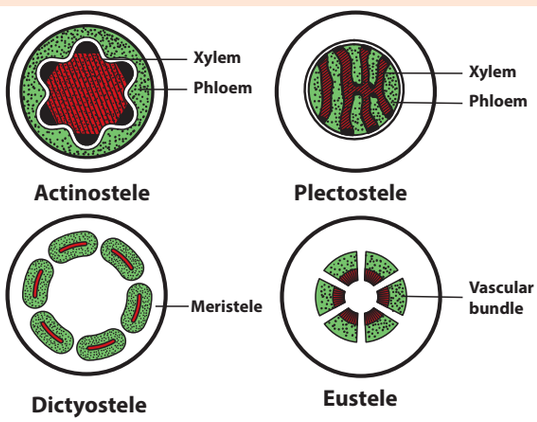


Figure 17: Types of Stele

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